School of Electronic Engineering

Using Machine Learning to Predict Scattering Parameters for Signal Integrity Purposes

Project Plan

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Contents

1 Introduction

Signal integrity is increasingly important in modern high-speed digital systems. As data rates on printed circuit boards (PCBs) increase, interconnect structures such as vias, traces, cavities and multilayer stack-ups introduce insertion loss, reflections, crosstalk and noise. These effects are commonly described using scattering parameters (S-parameters). Traditional physics-based simulation methods can model these responses accurately, but they are computationally expensive when many design variations must be explored [1, 2].

This project investigates whether machine learning can provide a practical surrogate-modelling approach for PCB interconnect analysis, and how different PCB dimensions and links in a topology affect the output and signal integrity. Using the open SI/PI-Database as the principal data source, the work will train models that learn the relationship between PCB design parameters and frequency-dependent signal-integrity responses derived from Touchstone data [3]. Physically meaningful constraints, such as signal insertion loss, interconnect passivity and causality would be considered while maintaining acceptable accuracy and physical consistency [4, 5].

1.1 Research Question and Objectives

Research question: Can a machine-learning forward model predict the complete broadband scattering matrix of a selected PCB interconnect topology, that is, all frequency-dependent S-parameters between all port pairs for the chosen multi-port structure, with sufficient accuracy for signal-integrity analysis while enforcing passivity and causality and reducing the cost of repeated electromagnetic simulation?

Project objectives:

- **Week 2:** Build a reproducible Python data pipeline to extract geometric and material parameters together with the complete Touchstone-based S-matrix for one selected topology from the SI/PI-Database.
- **Week 3:** Complete baseline scalar and frequency-conditioned prediction experiments to validate data parsing, feature scaling, train-validation-test splitting, and evaluation metrics.
- **Week 6:** Implement a broadband neural-network forward model that predicts the complete S-matrix for each design-frequency input, so that the response over the sampled frequency range can be reconstructed for the selected topology.
- **Week 8:** Incorporate physical-constraint enforcement, specifically passivity and causality, into the frequency-conditioned multi-output neural-network model and compare constrained and unconstrained variants.
- **Week 9:** Use the final physically constrained model to investigate how variation in each physical parameter of the interconnect affects the predicted S-matrix and derived signal-integrity behaviour.
- **14 August 2026 (End Date):** Complete a documented comparison between the proposed implementation and relevant prior broadband S-parameter neural-network approaches, using unseen test data, numerical error measures, physical-validity checks, parameter-variation analysis, and a clear discussion of limitations.

1.2 Project Scope

The project scope is intentionally focused so that the work remains feasible within the summer project while still addressing the core technical requirements of broadband S-parameter surrogate modelling.

Included in scope	**Out of scope / only discussed as future work**
One selected PCB topology from the SI/PI-Database as the primary modelling case	A universal model spanning all PCB topologies in the full SI/PI-Database
Broadband forward prediction of the complete S-matrix across frequency.	Inverse design or geometry optimisation from a desired S-parameter target.
Neural-network implementations that enforce or explicitly check passivity and causality.	A production-ready EDA workflow or deployment-ready software package.
Analysis of the effect of varying each physical design parameter on the predicted response.	Using only scalar frequency-conditioned prediction as the main project endpoint; this will be treated only as an early validation exercise.
Evaluation using held-out data, numerical error metrics, physical-validity measures, and representative plots of predicted versus reference responses.	Full industrial replacement of electromagnetic simulation workflows.

2 Project Design Approach and Methodology

The methodology will retain a short early validation stage, but its central task will be broadband forward modelling of the complete scattering matrix for one selected PCB topology from the SI/PI-Database. In line with the feedback, single-frequency prediction will be treated only as an early checkpoint to validate parsing, preprocessing, and evaluation by week 3. The main technical work will then focus on multi-frequency prediction of all S-parameters, with passivity and causality enforcement built into the final model, followed by an analysis of how variation in each physical design parameter affects the predicted response.

2.1 Data pipeline

The first task is to understand and structure the data correctly rather than rushing into model design. For each design-frequency sample, the inputs will be the relevant geometric and material parameters from *parameter.csv* together with the sampled frequency. The outputs will be derived from the corresponding Touchstone files, beginning with simple scalar targets for early validation and progressing to the complete complex scattering matrix at the same sampled frequency [3, 4, 6]. The work will begin with exploratory data analysis and parser validation. This includes checking parameter ranges, confirming mapping between parameter rows and Touchstone files, extracting target responses, and plotting representative S-parameter responses across frequency to confirm that the data behave as expected [3].

Original open-source datasets from SI/PI-Database project. Datasets from SI/PI-Database provided by Institut für Theoretische Elektrotechnik [3] will be used. Each SI/PI-Database dataset has a different PCB topology and may have different geometric and material parameters stored in a corresponding CSV file, *parameter.csv*, for example:

EPS	TAND	PITCH	TRACE_LEN	START	VIAR	ANTIPADR	TDIEL	DISTTL	TLWIDTH	SIMU_INDEX
3.8048	0.0142	53.0555	1833.2958	106.111	5.3119	7.3794	11.885	14.2377	4.4049	0
3.8682	0.0128	63.796	787.0598	127.592	6.0219	9.1892	13.342	18.2578	5.0949	1
4.2507	0.0079	73.7941	520.7679	147.588	3.5708	14.4234	13.938	31.3164	4.8946	10
4.2279	0.0167	61.3723	1877.4521	122.745	3.7778	14.3457	13.904	28.7384	4.9017	100

Figure 1: Example of design parameters stored in `parameter.csv`.

For the n-th simulation sample, the geometric and material parameters form the design-parameter vector

$$u^{(n)} = [u_1^{(n)}, u_2^{(n)}, \ldots, u_p^{(n)}]^T,$$

where p is the number of design features for the chosen topology. Here, $u_\ell^{(n)}$ denotes the ℓ-th design parameter of the n-th sample, for $\ell = 1, \ldots, p$, and superscript T denotes transpose.

In the selected topology, these may include parameters such as via pitch, via radius, antipad radius, cavity height, trace length, permittivity, and loss tangent. The exact parameter set depends on the chosen SI/PI-Database topology, since different datasets correspond to different PCB structures and therefore different geometric descriptions [3, 4].

For each parameter vector $u^{(n)}$, the corresponding simulation result is stored in a Touchstone file indexed by SIMU_INDEX. Therefore, the output for sample n is not a single scalar value, but a frequency-dependent matrix-valued response

$$S^{(n)}(\omega_k) = \begin{bmatrix} S_{11}^{(n)}(\omega_k) & \cdots & S_{1m}^{(n)}(\omega_k) \\ \vdots & \ddots & \vdots \\ S_{m1}^{(n)}(\omega_k) & \cdots & S_{mm}^{(n)}(\omega_k) \end{bmatrix} \in \mathbb{C}^{m \times m},$$

where m is the number of ports of the interconnect and $\omega = 2\pi f$ denotes angular frequency. Here, $S_{ij}^{(n)}(\omega_k)$ denotes the scattering response measured at port i when port j is excited, for the n-th design sample at the k-th sampled angular frequency point; $\mathbb{C}$ denotes the set of complex numbers. Thus, each raw simulation sample defines a broadband set of responses $\{S^{(n)}(\omega_k)\}_{k=1}^K$, and the supervised-learning dataset will be formed by evaluating this response at each sampled frequency.

To prepare the dataset for supervised learning, each simulation sample will be expanded over the sampled angular frequencies to form supervised-learning samples of the form

$$x^{(n,k)} = [u^{(n)}, \omega_k]^T, \quad k = 1, \ldots, K,$$

where $x^{(n,k)}$ denotes the input feature vector for the n-th design sample at the k-th sampled angular frequency point, $u^{(n)}$ is the design-parameter vector, and ω_k is included explicitly as part of the input. This design-frequency input representation will be used consistently for both the baseline modelling stage and the multi-output neural-network stage.

For the early baseline stage, the scalar target $y^{(n,k)}$ is defined according to the selected baseline modelling objective. For example, if the task is to predict the magnitude of one selected scattering parameter for port pair (i, j), then

$$y^{(n,k)} = \mid S_{ij}^{(n)}(\omega_k) \mid .$$

If insertion loss is used as an alternative scalar target for the early baseline, then

$$y^{(n,k)} = IL_{ij}^{(n)}(\omega_k) = -20 \log_{10} \mid S_{ij}^{(n)}(\omega_k) \mid .$$

For the main neural-network stage, the target for the same input $x^{(n,k)}$ is the complete complex S-matrix at that frequency:

$$Y^{(n,k)} = S^{(n)}(\omega_k) \in \mathbb{C}^{m \times m}.$$

The complete broadband response for design sample n is then reconstructed by evaluating the trained model over all sampled frequencies:

$$\{\widehat{S}^{(n)}(\omega_k)\}_{k=1}^{K}.$$

To avoid data leakage, all frequency points associated with a given design sample will remain in the same training, validation, or test split [4].

2.2 Baseline modelling

Once the data pipeline is validated, baseline non-neural regressors will be trained on scalar targets to establish reference performance and verify that the train-validation-test workflow is sound. The baseline models will include ridge regression and at least one tree-based regressor, such as random forest regression or gradient boosting regression. These models are widely used, relatively easy to interpret, and suitable for an initial benchmark. They provide a practical way to test whether the selected dataset contains useful predictive information before moving on to neural-network models. The input to each baseline model will stay consistent with the design-frequency representation defined above:

$$x^{(n,k)} = [u^{(n)}, \omega_k]^T.$$

Here, $u^{(n)}$ contains the geometric and material parameters of the selected PCB topology, while ω_k is the sampled angular frequency point. Depending on the dataset, the design parameters may include quantities such as via pitch, via radius, antipad radius, cavity height, trace length, permittivity, and loss tangent [4].

At this stage, the output will be a scalar signal-integrity target rather than the complete S-matrix. A suitable first target is the insertion loss for one selected transmission path at the input frequency ω_k:

$$y^{(n,k)} = IL_{ij}^{(n)}(\omega_k) = -20 \log_{10} \mid S_{ij}^{(n)}(\omega_k) \mid .$$

where i and j denote the selected port indices, with $S_{ij}^{(n)}(\omega_k)$ representing the response measured at port i when port j is excited for the n-th design sample at the k-th sampled angular frequency point. This baseline stage is intended only as an early validation exercise and should be completed by week 3. It keeps the first learning task simple while providing a practical check that data parsing, feature scaling, train-validation-test splitting, and evaluation metrics are working correctly. Insertion loss is also directly relevant to signal-integrity analysis, so it is a reasonable scalar baseline target [4].

Model performance will be measured on held-out validation and test sets. The main measures will be mean absolute error and root mean squared error. Here, N is the number of held-out design-frequency samples, $\widehat{y}^{(q)}$ denotes the predicted scalar target for held-out sample q, and $y^{(q)}$ denotes the corresponding reference target:

$$MAE = \frac{1}{N}\sum_{q=1}^{N} \mid \widehat{y}^{(q)} - y^{(q)} \mid$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{q=1}^{N} \left(\widehat{y}^{(q)} - y^{(q)}\right)^2}$$

The results will also be compared with a trivial mean predictor. This gives a basic reference point. The baseline stage will be considered successful if at least one non-neural model performs clearly better than that trivial predictor on unseen data, and if the validation and test errors remain reasonably close. If that is achieved, the project can move on to the neural-network stage with confidence that the data preparation, target extraction, and evaluation process are all working as intended.

2.3 Multi-output neural network

The main technical stage of the project is a multi-output neural network that predicts the complete S-matrix at a given design-frequency point. Unlike the baseline regressors, which predict only a single scalar target, this model will predict all S-parameter entries for the sampled frequency supplied as part of the input. The complete broadband response for an unseen design is reconstructed by evaluating the trained model over all frequency points in the sampled frequency grid. This makes the output useful for signal-integrity analysis, since insertion loss, return loss, and crosstalk can all be derived from the predicted S-matrix. This direction is consistent with the project requirement that prediction of the complete S-matrix should be a core part of the work.

The input to the model remains the same as in the baseline stage:

$$x^{(n,k)} = [u^{(n)}, \omega_k]^T.$$

For sample (n, k), the target is the complete complex S-matrix at the corresponding frequency:

$$Y^{(n,k)} = S^{(n)}(\omega_k) \in \mathbb{C}^{m\times m}.$$

Here, $S_{ij}^{(n)}(\omega_k)$ denotes the scattering response measured at port i when port j is excited. In compact form, the model can be written as

$$\widehat{S}^{(n)}(\omega_k) = f_\theta\left(u^{(n)}, \omega_k\right),$$

where f_θ denotes the neural network with learnable parameters θ. The hidden layers will learn the nonlinear relationship between PCB design parameters, frequency, and the S-parameter response. A feedforward fully connected neural network will be used as the initial architecture, with an input layer matching the number of design features plus the frequency feature, two or three hidden layers, and an output layer matching the chosen S-matrix representation at one frequency. If the real and imaginary parts of all S-parameter entries are stacked explicitly, the output dimension for each design-frequency sample is $2m^2$. Here, the factor 2 accounts for the stacked real and imaginary parts, and m^2 accounts for all port pairs. The frequency dimension is not included in the output layer size because frequency is an input feature rather than an output index.

Although the model receives one frequency point as part of its input, it is still a broadband model because the trained network is evaluated over the full sampled frequency grid for each unseen design:

$$\{\widehat{S}^{(n)}(\omega_k)\}_{k=1}^{K} = \{f_\theta\left(u^{(n)}, \omega_k\right)\}_{k=1}^{K}.$$

Model performance will be evaluated on unseen test designs using matrix-level MAE and RMSE across all predicted S-parameter entries and sampled frequencies. Here, N is the number of test designs, $\widehat{S}_{ij}^{(n)}(\omega_k)$ denotes the predicted scattering response, and $|\cdot|$ denotes the modulus of a complex quantity:

$$MAE = \frac{1}{Nm^2K}\sum_{n=1}^{N}\sum_{k=1}^{K}\sum_{i=1}^{m}\sum_{j=1}^{m} \mid \widehat{S}_{ij}^{(n)}(\omega_k) - S_{ij}^{(n)}(\omega_k) \mid .$$

$$RMSE = \sqrt{\frac{1}{Nm^2K}\sum_{n=1}^{N}\sum_{k=1}^{K}\sum_{i=1}^{m}\sum_{j=1}^{m} \mid \widehat{S}_{ij}^{(n)}(\omega_k) - S_{ij}^{(n)}(\omega_k) \mid^2}.$$

In addition, representative predicted-versus-reference responses will be plotted for selected port pairs on unseen test samples. This is important because a low aggregate error alone does not guarantee that the predicted broadband response is physically reasonable.

The implementation and output representation will be compared with the forward-model ideas of Akinwande et al. [6], which predict broadband complex S-parameters and incorporate a physically consistent layer to enforce passivity and causality. The model will be considered ready for the next stage when it achieves stable validation and test performance and reproduces the main trends of the broadband S-parameter responses on unseen samples.

2.4 Physical-consistency mechanism

For passivity, the predicted scattering matrix should satisfy $S(f_k)^H S(f_k) \preceq I$ for all sampled frequency points f_k, where H denotes conjugate transpose and I is the identity matrix. Equivalently, the largest singular value of the predicted matrix should satisfy $\sigma_{\max}(S(f_k)) \leq 1$, where $\sigma_{\max}(\cdot)$ denotes the largest singular value.

For causality, the predicted broadband response should correspond to a causal time-domain impulse response. A practical first implementation will be to use either a non-learnable correction or projection layer, or an equivalent penalty-based enforcement step, applied to the predicted S-matrix across frequency, informed by the forward-model literature [6]. Non-negative insertion loss may still be reported as a derived physical sanity check, but it will no longer be the main physical constraint.

The input and output of the model will remain the same. The difference is that the training objective, or possibly the output design, will be modified to encourage physically valid predictions. The total loss can be written as

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda\,\mathcal{L}_{\text{phys}},$$

where $\mathcal{L}_{\text{data}}$ is the ordinary prediction loss and $\mathcal{L}_{\text{phys}}$ penalises non-physical outputs. A practical choice is to penalise violations of the passivity condition using the largest singular value of the predicted S-matrix:

$$\mathcal{L}_{\text{phys}} = \frac{1}{NK} \sum_{n=1}^{N} \sum_{k=1}^{K} \max\left(0, \sigma_{\max}\left(\widehat{S}^{(n)}\left(\omega_k\right)\right) - 1\right).$$

This follows the general idea of physics-enforced modelling, but is adapted here to enforce passivity of the full S-matrix rather than only insertion loss. In addition, causality may be enforced through a minimum-phase or equivalent frequency-domain consistency mechanism applied across the predicted broadband response [4, 6].

Performance will still be measured using matrix-level MAE and RMSE across all S-parameter entries and sampled frequencies on held-out validation and test sets. This stage will also add a physical-validity measure: the proportion of design-frequency pairs for which passivity is violated,

$$r_{viol} = \frac{1}{NK} \sum_{n=1}^{N} \sum_{k=1}^{K} \mathbf{1}\left(\sigma_{\max}\left(\widehat{S}^{(n)}\left(\omega_k\right)\right) > 1\right),$$

where $\mathbf{1}(\cdot)$ is the indicator function. The constrained model will be considered successful if it reduces or removes these violations while maintaining prediction accuracy close to that of the unconstrained multi-output model.

Dataset selection	**Parsing and target extraction**	**Baseline models**	**Multi-output NN**	**Physical checks and evaluation**
Select one primary topology and freeze dataset version	Build design-frequency samples from `parameter.csv` and Touchstone files	Validate pipeline with simple scalar targets	Predict complete S-matrix for each design-frequency input	Check accuracy, generalization and physical plausibility

After the final constrained model has been trained, each physical design parameter will be varied in turn while the remaining parameters are held at nominal values or representative settings. The resulting changes in selected S-parameters, insertion loss, return loss, and crosstalk across frequency will be plotted and discussed. This makes variation of each physical parameter a core output of the final neural-network implementation rather than future work.

3 Workplan

The project is scheduled to run from 11 May 2026 to 31 July 2026 for core technical work, with the period from 3 August 2026 to 14 August 2026 reserved for contingency, dissertation finalisation, proofreading, formatting, and submission preparation.

Milestone	**Weeks**	**Main tasks**	**Deliverables**

Milestone 1 - Data pipeline and problem confirmation	Weeks 1-2 (11 May - 24 May)	Freeze the primary dataset; implement parsing of `parameter.csv` and Touchstone files; confirm target extraction; produce sanity-check plots and notes on the selected topology.	Reproducible data-preparation scripts and exploratory analysis pack.
Milestone 2 - Baselines and scalar targets	Weeks 3-4 (25 May - 7 Jun)	Train non-neural baseline regressors and simple scalar-prediction models; validate splits, scaling and evaluation metrics; compare alternative targets for usefulness.	Baseline results table, scalar-target error plots and model-selection note.
Milestone 3 - Trainable frequency-conditioned complete S-matrix prototype with first held-out results	Weeks 5-7 (8 Jun - 28 Jun)	Implement the first multi-output neural network for frequency-conditioned S-matrix prediction; refine feature representation and S-matrix output representation; begin hyperparameter tuning.	Trainable multi-output prototype with first held-out results.
Milestone 4 - Model refinement and physical consistency	Weeks 8-9 (29 Jun - 12 Jul)	Improve architecture and training stability; add physical-consistency mechanism; run ablation comparisons between unconstrained and constrained variants; document limitations.	Refined target model, comparison study and physical-validity checks.
Milestone 5 - Final evaluation, writing and packaging	Weeks 10-12 (13 Jul - 2 Aug)	Consolidate experiments; prepare final plots and tables; complete the main written report and supporting materials; attempt limited stretch work only if core objectives are already secure.	Core report content, reproducible code package and presentation-ready summary material.
Contingency and submission buffer	Weeks 13-14 (3 Aug - 14 Aug)	Use the reserved period for proofreading, reference checks, formatting fixes, supervisor feedback incorporation, final packaging and submission preparation.	Submission-ready dissertation package with contingency preserved.

3.1 Resource management

The project will use the datasets already obtained under the existing agreement, together with a local Python-based development environment using TensorFlow/Keras, Jupyter Notebook and standard scientific Python libraries. No specialist hardware purchases are required. The main resources are computing time, disciplined experiment logging, and continuous supervision feedback on scope and interpretation of results.

3.2 Risk assessment and mitigations

Risk	Impact	Likelihood	Mitigation
Dataset-topology dependence limits generalization	Model may only work well on the selected topology	High	Treat one topology as the core case, state generalization limits clearly, and frame broader topology transfer as future work.
High output dimensionality causes unstable training or overfitting	Poor generalization on unseen designs	Medium	Use the progressive formulation strategy, regularization, validation-based model selection, and begin with simpler scalar or selected-port representations before progressing to full S-matrix prediction.
Non-physical predictions	Model becomes less credible for SI analysis	Medium	Incorporate a physical-consistency mechanism and make physical validity part of model evaluation rather than an afterthought.
Time compression during the summer window	Core deliverables may be incomplete	Medium	Prioritise milestone completion, write continuously throughout the project, and keep stretch work conditional on early success of the target formulation.
Tooling or environment instability	Experiment delays and reproducibility issues	Low	Use version-controlled notebooks/scripts, freeze package versions where possible, and maintain simple reproducible pipelines.

4 Proposed outcomes

The primary expected outcome is a reproducible machine-learning workflow that predicts signal-integrity-relevant responses from PCB interconnect design parameters more quickly than repeated physics-based simulation [3, 2]. The main technical output is expected to be a multi-output neural-network model that predicts the complete S-matrix for each design-frequency input, from which the broadband S-matrix response over frequency is reconstructed for a selected PCB topology. Insertion loss, return loss, and crosstalk are derived from the predicted S-matrix rather than directly predicted [4, 5].

The research contribution is expected to be modest but meaningful at MEng level. First, it will demonstrate a structured pathway from dataset understanding to usable surrogate modelling for signal integrity. Second, it will evaluate the trade-off between prediction accuracy and physical plausibility in this domain. Third, it will provide a clear discussion of where the approach is useful, where topology dependence limits generalization, and which aspects should be treated as future work rather than overclaimed results.

Comparison with the state of the art will be made through the following measurable criteria:

- Baseline versus target-model comparison on held-out test data using mean-squared error and matrix-level error measures across S-parameter entries and frequencies.
- Representative predicted versus ground-truth plots over frequency for unseen samples.
- Evidence that the final model behaves in a physically meaningful way for the chosen target response.
- A clear account of limitations, such as topology dependence, dataset coverage and the boundaries of model applicability.

4.1 Novel aspects of the proposed work

The novelty of the proposed work does not lie simply in predicting a frequency-dependent signal-integrity response, since prior work has already addressed insertion-loss modelling over frequency [4]. Instead, the contribution is the development of a staged, reproducible, and physics-aware surrogate-modelling workflow for the selected PCB topology using the SI/PI dataset [3]. The work will compare progressively more capable formulations, from baseline regression models to frequency-conditioned complete S-matrix prediction, and will assess them using both numerical accuracy and physical-consistency criteria relevant to signal-integrity analysis.

In addition, the project adopts an iterative and progressive development methodology informed by software engineering practice. Rather than attempting a complex end-to-end solution immediately, the system will be evolved in controlled stages, with explicit performance targets required before progression to the next model class. This approach improves traceability, reduces technical risk, and strengthens the reproducibility of the final dissertation outcomes.

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5 Appendix

A1. Risk assessment

The principal project risks are dataset-topology dependence, unstable training due to multi-output S-matrix prediction, and the possibility of non-physical predictions. These have been addressed in the main workplan by adopting a single-topology focus, progressive model complexity, and explicit physical-consistency checks.

A.2 Components to be ordered

No hardware components are expected to be ordered for this project. The work is software-based and uses already available datasets and computing resources.